

Austin J. Baird, PhD

RESEARCH ASSISTANT PROFESSOR · DEPARTMENT OF SURGERY | DIVISION OF HEALTHCARE SIMULATION SCIENCES

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Education

University of California, Santa Cruz

BA IN MATHEMATICS

Santa Cruz, CA

September 2004 - June 2008

- *Honors Thesis:* Modeling Native California Grassland Populations

University of North Carolina, Chapel Hill

PHD IN APPLIED MATHEMATICS

Chapel Hill, NC

August 2009 - August 2014

- *Thesis:* Modeling Valveless Pumping Mechanisms, [link](#)
- *Advisor:* Dr. Laura Miller, *Committee:* Forest, M. Gregory, Adalsteinsson, David, White, Brian, Mucha, Peter, Kier, William M.

Postgraduate Training

NONE

Faculty Positions Held

Duke University

VISITING ASSISTANT PROFESSOR

Durham, NC

August 2014 - July 2015

University of Washington

RESEARCH ASSISTANT PROFESSOR

Seattle, Wa

December 2021 - Present

Hospital Positions Held

NONE

Honors

University of North Carolina, Chapel Hill

FUTURE FACULTY FELLOW

Chapel Hill, NC

June 2012

University of North Carolina, Chapel Hill

J. BURTON LINKER AWARD IN RECOGNITION OF EXCELLENCE IN TEACHING

Chapel Hill, NC

March 2013

University of North Carolina, Chapel Hill
GRADUATE ASSISTANCE IN AREAS OF NATIONAL NEED FELLOW

Chapel Hill, NC
Fall 2013

Applied Research Associates, Inc.
ABOVE AND BEYOND THE CALL OF DUTY AWARD

Raleigh, NC
October 2016

Department of Defense
FEDERAL IT INNOVATIONS AWARD

Washington D.C.
August 2017

Applied Research Associates, Inc.
DISTINGUISHED MEMBER OF THE TECHNICAL STAFF

Raleigh, NC
June 2018

Applied Research Associates, Inc.
TECHNICAL EXCELLENCE AWARD

Raleigh, NC
September 2019

Board Certifications

NONE

Current Licenses to Practice

NONE

Diversity and Equity Inclusion Activities

Graduate Mathematics Association at UNC, Chapel Hill
SONIA KOVALESKY DAY WORKSHOP LEAD

Chapel Hill, NC
May 2012

Graduate Mathematics Association at UNC, Chapel Hill
MINORITY SCIENTIST MENTOR

Chapel Hill, NC
May 2013

WISE, NC State University
WOMEN IN SCIENCE AND ENGINEERING SPEAKER

Raleigh, NC
March 2017

Affiliations

Society for Simulation in Healthcare

LEADERSHIP GROUP ON SIMULATION AND MODELING

2016 - Present

IEEE

ENGINEERING IN MEDICINE AND BIOLOGY

2018 - Present

NIH

INTER-AGENCY MODELING AND ANALYSIS GROUP

2019 - Present

Teaching Experience

Champion School

COMPUTER SCIENCE, MATH, AND PHYSICS MIDDLE SCHOOL INSTRUCTOR

San Jose, CA

August 2008

- Full course responsibility

University of North Carolina, Chapel Hill, Math 190

CHAOS AND POPULATION DYNAMICS

Chapel Hill, NC

Fall 2012

- Full course responsibility

University of North Carolina, Chapel Hill, Math 290

APPLIED MATHEMATICAL METHODS (COMPUTATIONAL LAB)

Chapel Hill, NC

Spring 2013

- Full course responsibility
- *Course Materials:* [link](#)

Duke University, Math 353

INTRODUCTION TO ORDINARY AND PARTIAL DIFFERENTIAL EQUATIONS FOR ENGINEERS

Durham, NC

Fall 2014

- Full course responsibility

Duke University, Math 353

INTRODUCTION TO ORDINARY AND PARTIAL DIFFERENTIAL EQUATIONS FOR ENGINEERS

Durham, NC

Spring 2015

- Full course responsibility

Mentees

APPLIED RESEARCH ASSOCIATES

Raleigh, NC

Spring 2016 - October 2021

- Laura Oelsner
Dates: May 2018 - August 2019
Current Position: Becton Dickinson Biomedical Engineer
- Tim Hoer
Dates: May 2018 - August 2018
Current Position: Fuel Cell System Engineer
- Rishi Das
Dates: May 2020 - August 2020
Current Position: Harvard Medical School Engineering Internship

Seattle, WA
Oct 2022 - Present

UNIVERSITY OF WASHINGTON

- Namrata Harish
Dates: Oct 2022 - Dec 2023
Current Position: BioEngineering Student, University of Washington
- Jake Flynn, Megan Luu, Sanika Joshi
Dates: Sep 2022 - May 2023
Current Position: BioEngineering Student, University of Washington
- Issac Kim
Sep 2023 - 2024
Current Position: BioEngineering Student Capstone, University of Washington
- Jake Kelly, Rylie Darlington, Riley Turbek
Dates: Sep 2023 - Present
Current Position: BioEngineering Student Capstone, University of Washington
- Namrata Harish, Nitya Lagadapati, Natalie Heitkamp
Dates: Sep 2024 - Present
Current Position: BioEngineering Student Capstone, University of Washington
- Shu-Yi Yeh
Dates: Sep 2024 - Present
Current Position: Masters Student BioEngineering
- Issac Kim
Dates: Sep 2024 - Present
Current Position: Masters Student BioEngineering

Editorial Responsibilities

Frontiers in Physiology: Computational Physiology and Medicine

REVIEW EDITOR

June 2022 - Present

Journal of Open Source Software

REVIEW EDITOR

Jan 2024 - Present

National Responsibilities

Society of Simulation in Healthcare

CHAIR: HEALTHCARE SYSTEMS MODELING AFFINITY GROUP

March 2023 - Present

Local Responsibilities

University of Washington

FACULTY SENATE-COUNCIL ON CYBER SECURITY

Seattle, Wa
September 2023 - Present

Funding

Defense Health Agency

BIOGEARS RESEARCH AWARD

- *Role:* Principal Investigator
- *Amount:* \$7,800,000
- *Contract(s):* W81XWH-13-2-0068,

Defense Health Agency

BIOGEARS FOLLOW-ON RESEARCH AWARD

- *Role:* Principal Investigator
- *Amount:* \$1,900,000
- *Contract(s):* W81XWH-17-C-0172

Army Research Labs

FAST COMPUTATIONAL SIMULATIONS OF TRAUMATIC BRAIN INJURY

- *Role:* Principal Investigator
- *Amount:* \$353,000
- *Contract(s):* W911NF-17-1-0572

Defense Health Agency

BURN CARE: VIRTUAL PATIENT APPLICATION TO TRAIN THERMAL INJURY

- *Role:* Principal Investigator
- *Amount:* \$2,100,000
- *Contract(s):* W911NF-18-C-0037

Defense Health Agency

SUSTAIN: PROLONGED FIELD CARE TRAINING FRAMEWORK

- *Role:* Lead Physiology Modeler and Proposal Manager
- *Amount:* \$2,200,000
- *Contract(s):* W81XWH-18-C-0169

CyberPatient

CYBERPATIENT: TECHNICAL CONSULTATIONS

- *Role:* PI
- *Amount:* \$23,910

Regis University

OVERWATCH AI

- *Role:* Co-PI
- *Amount:* \$68,782.59

VCom3D

AFTTS

- *Role:* Physiology Modeling Services
- *Amount:* \$100,000.00

VCom3D

EXTREME-PHYSIOLOGICAL MODELS AND TRAINING

- *Role:* PI
- *Amount:* \$160,000.00

Publications

FIRST SECTION: PEER REVIEWED ARTICLES

[1] Modeling Valveless Pumping Mechanisms

A. BAIRD

College of Arts and Sciences, Department of Mathematics (2014)

[2] Neuromechanical Pumping: Boundary Flexibility and Traveling Depolarization Waves Drive Flow Within Valveless, Tubular Hearts

Fredrick, MD

August 2013 - August 2018

Fredrick, MD

August 2017 - August 2020

Raleigh, NC

January 2017 - December 2020

Orlando, FL

January 2018 - Dec 2021

Fredrick, MD

March 2018 - Dec 2021

Vancouver, BC

Oct 2022 - Jan 2023

Denver, CO

March 2023 - Present

Orlando, FL

September 2023 - Present

Orlando, FL

September 2023 - Present

A. BAIRD, L. WALDROP, L. MILLER

Japan Journal of Industrial and Applied Mathematics 32.3 (2015) pp. 829–846 Springer

- [3] **A Mathematical Model and MATLAB Code for Muscle–Fluid–Structure Simulations**
N. A. BATTISTA, A. J. BAIRD, L. A. MILLER
Integrative and comparative biology 55.5 (2015) pp. 901–911 Oxford University Press
- [4] **A Full-Body Model of Burn Pathophysiology and Treatment Using the BioGears Engine**
M. MCDANIEL, A. BAIRD
2019 41st Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC) (2019) pp. 261–264 IEEE
- [5] **Open Source Pharmacokinetic/Pharmacodynamic Framework: Tutorial on the BioGears Engine**
M. MCDANIEL, J. CARTER, J. M. KELLER, S. A. WHITE, A. BAIRD
CPT: pharmacometrics & systems pharmacology 8.1 (2019) pp. 12–25 Wiley Online Library
- [6] **A Whole-Body Mathematical Model of Sepsis Progression and Treatment Designed in the BioGears Physiology Engine**
M. MCDANIEL, J. KELLER, S. WHITE, A. BAIRD
Frontiers in physiology 10 (2019) p. 1321 Frontiers
- [7] **BioGears: A C++ library for whole body physiology simulations**
A. BAIRD, M. MCDANIEL, S. A. WHITE, N. TATUM, L. MARIN
Journal of Open Source Software 5.56 (2020) p. 2645
- [8] **BurnCare tablet trainer to enhance burn injury care and treatment**
A. BAIRD, M. SERIO-MELVIN, M. HACKETT, M. CLOVER, M. MCDANIEL, M. ROWLAND, A. WILLIAMS, B. WILSON
BMC Emergency Medicine 20.1 (2020) pp. 1–10 BioMed Central
- [9] **Towards in Silico Clinical Trials for Sepsis Patients**
B. AUSTIN, M. MATTHEW, K. JONATHAN
SIAM.org (2021)
- [10] **A multiscale computational model of angiogenesis after traumatic brain injury, investigating the role location plays in volumetric recovery**
A. BAIRD, L. OELSNER, C. FISHER, M. WITTE, M. HUYNH
Mathematical Biosciences and Engineering 18.4 (2021) pp. 3227–3257
- [11] **Whole body physiology model to simulate respiratory depression of fentanyl and associated naloxone reversal**
A. BAIRD, S. A. WHITE, R. DAS, N. TATUM, E. K. BISGAARD
Communications Medicine 4.1 (2024) p. 114 Nature Publishing Group UK London
- [12] **Increasing Realism and Variety of Virtual Patient Dialogues for Prenatal Counseling Education Through a Novel Application of ChatGPT: Exploratory Observational Study**
G. M, B. A, S. T, J. J. DEBROUX, T. BARTLETT, M. KRICK, U. R
JMIR Med Educ 10 (2024)
- [13] **A Whole-Body Physiology Model to Investigate Respiratory Function During Exercise Across Different Age Cohorts**
B. AUSTIN, U. R. A, W. S. A, G. MEGAN, S. T. L
Journal of Respiration 5.1 (2025) p. 1 MDPI

SECOND SECTION: COLLABORATIVE AUTHORSHIP

None

THIRD SECTION: MEDEDPORTAL

None

FOURTH SECTION: BOOK CHAPTER

- [1] **Numerical Study of Scaling Effects in Peristalsis and Dynamic Suction Pumping**

A. BAIRD, T. KING, L. MILLER

Contemp. Math 628 (2014) pp. 129–148

FIFTH SECTION: PUBLISHED BOOKS, VIDEOS, SOFTWARE, ECT.

- [1] **ajbaird/TBISimulator: Alpha 1.0.1 Windows executable and data**

A. BAIRD

(Sept. 2020) Zenodo

- [2] **BioGears: A C++ library for whole body physiology simulations**

A. BAIRD, M. MCDANIEL, S. A. WHITE, N. TATUM, L. MARIN

(Dec. 2020) Zenodo

- [3] **BurnCare Physiology Data: Burn with Resuscitation**

N. TATUM, A. BAIRD, S. WHITE, M. CLOVER, M. HACKETT

(June 2021) Zenodo

- [4] **BurnCare Physiology Data: Burn with Resuscitation and Escharotomy**

N. TATUM, A. BAIRD, S. WHITE, M. CLOVER, M. HACKETT

(Aug. 2021) Zenodo

- [5] **BurnCare Physiology Data: Burn without Intervention**

N. TATUM, A. BAIRD, S. WHITE, M. CLOVER, M. HACKETT

(Feb. 2021) Zenodo

- [6] **Whole-body physiology model to simulate respiratory depression of fentanyl and associated naloxone reversal [Dataset]**

A. BAIRD, E. BISGAARD

(Feb. 2024) Dryad

SIXTH SECTION: OTHER PUBLICATIONS

- [1] **Electro-Dynamic Suction Pumping at Small Scales**

A. BAIRD, L. MILLER

APS Division of Fluid Dynamics Meeting Abstracts, 2013

SEVENTH SECTION: SUBMITTED MANUSCRIPTS

- [1] **Automated Trauma Treatment Using Reinforcement Learning and a Dynamic Whole-Body Physiology Model**

A. BAIRD, M. BUSCH, D. HART

IEEE - Journal of Biomedical and Health Informatics (2025)

- [2] **A Model of Anaerobic Tissue Perfusion During Trauma – Lactate Trajectory Curvature Can Determine Recovery**

A. BAIRD, S. WHITE, E. BISGAARD

PLOS Computational Biology (2025)

- [3] **Gender Specific Differences in Advanced Airway Trauma: Open-Source Physiology Engines Can Fill the Gap and Reduce Cost Barriers**

A. BAIRD, S. A. WHITE, E. K. BISGAARD, I. BENTOV, E. TIMM, R. LEUSCHKE, V. ROACH, R. SWEET, D. HANANEL
JMIR - Medical Education (2025)

[4] **Physiology-Driven Reversible Dynamic Synthetic Tissue Model for Responsive Moulage in Infected Wound Training**

A. GONG, A. DESCHMIDT, A. BAIRD, J. GONZAGA, S.-W. YAU, R. TOEPFER, C. PARSEY, R. LEUSCHKE, J. NORFLEET, R. SWEET
JMIR - Medical Education (2025)

[5] **Physical Findings Database for Simulation Education – An Open-Source Resource for High Fidelity Teaching Technologies**

D. HANANEL, A. BAIRD, L. PONIATOWSKI, J. KELLER, R. SWEET
Simulation in Healthcare (2025)

FINAL SECTION: LIST OF ABSTRACTS

A FULL-BODY MODEL OF BURN PATHOPHYSIOLOGY AND TREATMENT USING THE BIOGEARS ENGINE

- We have created a model of systemic burn pathophysiology by incorporating a mathematical model of acute inflammation within the BioGears Engine. This model produces outputs consistent with burns of varying severities and leverages existing BioGears functionality to simulate the effect of treatment on virtual patient outcome. The model performs well for standard resuscitation scenarios and we thus expect it to be useful for educational and training purposes.

A WHOLE-BODY MATHEMATICAL MODEL OF SEPSIS PROGRESSION AND TREATMENT DESIGNED IN THE BIOGEARS PHYSIOLOGY ENGINE

- Sepsis is a debilitating condition associated with a high mortality rate that greatly strains hospital resources. Though advances have been made in improving sepsis diagnosis and treatment, our understanding of the disease is far from complete. Mathematical modeling of sepsis has the potential to explore underlying biological mechanisms and patient phenotypes that contribute to variability in septic patient outcomes. We developed a comprehensive, whole-body mathematical model of sepsis pathophysiology using the BioGears Engine, a robust open-source virtual human modeling project. We describe the development of a sepsis model and the physiologic response within the BioGears framework. We then define and simulate scenarios that compare sepsis treatment regimens. As such, we demonstrate the utility of this model as a tool to augment sepsis research and as a training platform to educate medical staff.

OPEN SOURCE PHARMACOKINETIC/PHARMACODYNAMIC FRAMEWORK: TUTORIAL ON THE BIOGEARS ENGINE

- BioGears is an open-source, lumped parameter, full-body human physiology engine. Its purpose is to provide realistic and comprehensive simulations for medical training, research, and education. BioGears incorporates a physiologically based pharmacokinetic/pharmacodynamic (PK/PD) model that is designed to be applicable to a diversity of drug classes and patients and is extensible to future drugs. In addition, BioGears also supports drug interactions with various patient insults and interventions allowing for a realistic research framework and accurate dose-patient responses. This tutorial will demonstrate how the generic BioGears PK/PD model can be extended to a new substance for prediction of drug administration outcomes.

A MULTISCALE COMPUTATIONAL MODEL OF ANGIOGENESIS AFTER TRAUMATIC BRAIN INJURY, INVESTIGATING THE ROLE LOCATION PLAYS IN VOLUMETRIC RECOVERY

- Vascular endothelial growth factor (VEGF) is a key protein involved in the process of angiogenesis. VEGF is of particular interest after a traumatic brain injury (TBI), as it re-establishes the cerebral vascular network in effort to allow for proper cerebral blood flow and thereby oxygenation of damaged brain tissue. For this reason, angiogenesis is critical in the progression and recovery of TBI patients in the days and weeks post injury. Although well established experimental work has led to advances in our understanding of TBI and the progression of angiogenesis, many constraints still exist with existing methods, especially when considering patient progression in the days following injury. To better understand the healing process on the proposed time scales, we develop a computational model that quickly simulates vessel growth and recovery by coupling VEGF and its interactions with its associated receptors to a physiologically inspired fractal model of the microvascular re-growth. We use this model to clarify the role that diffusivity, receptor kinetics and location of the TBI play in overall blood volume restoration in the weeks post injury and show that proper therapeutic angiogenesis, or vasculogenic therapies, could speed recovery of the patient as a function of the location of injury.

BIOGEARS: A C++ LIBRARY FOR WHOLE BODY PHYSIOLOGY SIMULATIONS

- BioGears is an open source, extensible human physiology computational engine that is designed to enhance medical education, research, and training technologies. BioGears is primarily written in C++ and uses an electric circuit analog to characterize the fluid dynamics of the cardiopulmonary system. As medical training requirements become more complex, there is a need to supplement traditional simulators with physiology simulations. To this end, BioGears provides an extensive number of validated injury models and related interventions that may be applied to the simulated patient. In addition, BioGears compiled libraries may be used for computational medical research to construct in-silico clinical trials related to patient treatment and outcomes. Variable patient inputs support diversity and specification in a given application. The engine can be used standalone or integrated with simulators, sensor interfaces, and models of all fidelities. The Library, and all associated projects, are published under the Apache 2.0 license and are made available through the public GitHub repository. BioGears aims to lower the barrier to create complex physiological simulations for a variety of uses and requirements

BURN CARE TABLET TRAINER TO ENHANCE BURN INJURY CARE AND TREATMENT

- Applied Research Associates (ARA) and the United States Army Institute of Surgical Research (USAISR) have been developing a tablet-based simulation environment for burn wound assessment and burn shock resuscitation. This application aims to supplement the current gold standard in burn care education, the Advanced Burn Life Support (ABLS) curriculum. Subject matter experts validate total body surface area (TBSA) identification and analysis and show that the visual fidelity of the tablet virtual patients is consistent with real life thermal injuries. We show this by noting that the error between their burn mapping and the actual patient burns was sufficiently less than that of a random sample population. Statistical analysis is used to confirm this hypothesis. In addition a full body physiology model developed for this project is detailed. Physiological results, and responses to standard care treatment, are detailed and validated. Future updates will include training modules that leverage this model. We have created an accurate, whole-body model of burn TBSA training experience in Unreal 4 on a mobile platform, provided for free to the medical community. We hope to provide learners with more a realistic experience and with rapid feedback as they practice patient assessment, intervention, and reassessment.

WHOLE BODY PBPK MODEL OF NASAL NALOXONE ADMINISTRATION TO MEASURE REPEAT DOSING

- Opioid use in the United States and abroad is an endemic part of culture with yearly increases in overdose rates and deaths. As rates of overdose incidence increases, the use of the safe and effective reversal agent, Naloxone, in the form of a nasal rescue spray is being fielded and used by emergency medical technicians (EMTs) at a greater and greater rate. Despite advances in deployment of these rescue products, deaths are continuing to increase. There is evidence that repeated dosing of a Naloxone nasal spray (such as Narcan) is becoming more common due to the amount and type of opiate being abused. Although there is strong evidence of the benefits of Naloxone related to opioid reversals there are still not repeated dosing guidelines, as a function of opiate and amount the patient has taken. Goal directed dosing is promising, where respiratory markers are being used as an indication of the patient recovery but require time and understanding by the EMT. We construct a whole-body model of the pharmacokinetics and dynamics of opiates (Morphine and Fentanyl) on respiratory depression. We then construct a model of nasal deposition and administration of naloxone to investigate repeat dosing requirements for large overdoses and analyze how the two opiates may influence these requirements. We show that naloxone is highly effective at reversing respiratory symptoms of the patient and recommend dosing requirements as a function of opiate and amount administered. By designing the model to include circulation and respiration we investigate physiological markers that may be used in goal directed therapy rescue treatments.

INCREASING REALISM AND VARIETY OF VIRTUAL PATIENT DIALOGUES FOR PRENATAL COUNSELING

EDUCATION THROUGH A NOVEL APPLICATION OF CHATGPT: EXPLORATORY OBSERVATIONAL STUDY

- Using virtual patients, facilitated by natural language processing, provides a valuable educational experience for learners. Generating a large, varied sample of realistic and appropriate responses for virtual patients is challenging. Artificial intelligence (AI) programs can be a viable source for these responses, but their utility for this purpose has not been explored.

A WHOLE-BODY PHYSIOLOGY MODEL TO INVESTIGATE RESPIRATORY FUNCTION DURING EXERCISE ACROSS

DIFFERENT AGE COHORTS

- There is a limitation in understanding complete cardiopulmonary function during moments of exhaustive exercise due to invasive measurement techniques. We asked how cardiopulmonary function is perturbed during times of exercise in different age cohorts: 6–11 and 11–18. We sought to broadly analyze how age and oxidative stress during exercise differ across age cohorts. Methods: We present a whole-body physiology model that connects the nervous, cardiovascular, respiratory, and oxygen transport and binding systems. We connect these models using a lumped parameter representation of the cardiovascular and respiratory systems. Results: we observe distinct age-related difference in physiological response to exercise. These responses consist of respiratory, cardiovascular, and nervous system perturbations that are distinct across these age groups.

Invited Presentations

Integrative and Mechanical Physiology Group

PUMPING MECHANISMS AND THEIR SCALING EFFECTS IN TUBULAR HEARTS

Chapel Hill, NC

September 2012

Society of Integrative and Comparative Biology

SCALING AND PUMPING IN TUBULAR HEARTS

San Francisco, CA

January 2013

Integrative and Mechanical Physiology Group

MODELING NEURONS

Chapel Hill, NC

March 2013

Society of Mathematical Biology

ELECTRO-DYNAMIC SUCTION PUMPING AT SMALL SCALES

Osaka, Japan

August 2014

Duke Interdisciplinary Discussion Course

MOVING FLUID IN TUBES

Durham, NC

October, 2014

Experimental Biology

IMPLICATIONS OF INCREASE RENAL VENOUS PRESSURE FOR RENAL HEMODYNAMIC AND REABSORPTIVE
FUNCTION STUDIED BY A MATHEMATICAL MODEL OF THE KIDNEY

Boston, MA

March 2015

Chemical and Biological Defense Science and Technology Conference

BIOGEARS: SIMULATING WHOLE-BODY RESPONSE TO CHEMICAL EXPOSURE

Long Beach, CA

November 2017

International Meeting on Simulation in Healthcare

AN IN-SILICO WHOLE-BODY FRAMEWORK TO SIMULATE KINETICS AND DYNAMICS OF PHARMACEUTICALS
AND ASSOCIATED REVERSAL AGENTS

Los Angeles, CA

January 2018

Virtual Physiological Human Conference

AN IN-SILICO WHOLE-BODY FRAMEWORK TO SIMULATE KINETICS AND DYNAMICS OF PHARMACEUTICALS
AND ASSOCIATED REVERSAL AGENTS

Zaragoza, Spain

September 2018

Department of Defense Working Group on Computational Modeling of Human Lethality, Injury, and Impairment from Blast-Related Threats

BIOGEARS HUMAN PHYSIOLOGY ENGINE

Arlington, VA

February 2019

American College of Surgeons Simulation Summit

BIOGEARS: A FRAMEWORK FOR MULTISCALE PHYSIOLOGY MODELING

Chicago, IL

March 2019

Society for Simulation in Healthcare

BIOGEARS MODEL TO SIMULATE PATIENT RESPONSES TO SEPSIS

Raleigh, NC

March 2019

BioGears Conference

"BIOGEARS DRUG MODELING OVERVIEW"

Raleigh, NC

March 2020

- *Conference Talks:* [link](#)

University of Arizona Modeling and Computation Seminar

"A WHOLE-BODY PHYSIOLOGICAL MODEL OF SEPSIS AND ASSOCIATED TREATMENTS, DESIGNED IN THE BIOGEARS ENGINE"

Tucson, AZ

March 2021

University of Arizona Modeling and Computation Seminar

"A WHOLE-BODY PHYSIOLOGICAL MODEL OF SEPSIS AND ASSOCIATED TREATMENTS, DESIGNED IN THE BIOGEARS ENGINE"

Tucson, AZ

March 2021

CyberPatient Technical Interest Meeting

"BIOGEARS: PHYSIOLOGICAL MODELING AND ENGINE CONSTRUCTION FOR HIGH FIDELITY SIMULATION TRAINING"

Virtual

Jan 2022

University of Washington Biological Engineering Design Seminar

"ENGINEERING DESIGN WITH BIOLOGICAL PURPOSE"

Seattle, WA

April 2022

University of Washington Biological Engineering Seminar

"BIOGEARS: WHOLE-BODY PHYSIOLOGICAL MODEL TO SUPPORT HEALTHCARE SIMULATION, RESEARCH AND TRAINING"

Seattle, WA

April 2022

Society of Industrial and Applied Mathematics: Life Sciences Conference

"MODELING THE WHOLE-BODY RESPONSE TO INFECTION AND ASSOCIATED ACUTE INFLAMMATION, INVESTIGATING CLINICAL TREATMENTS AND OUTCOMES"

Pittsburgh, PA

July 2022

ACS-Simulation Summit

"GENDER SPECIFIC DIFFERENCES IN ADVANCED AIRWAY TRAUMA: PHYSIOLOGY ENGINES CAN FILL THE GAP"

Chicago, IL

Feb 2023

NIH-Multiscale Modeling Group

"PBPK MODEL OF NASAL ADMINISTRATION OF NALOXONE TO MEASURE REPEAT DOSING"

Bethesda, MD

June 2023

Healthcare Systems Modeling and Simulation Quarterly Meeting

"BIOGEARS: WHOLE-BODY PHYSIOLOGICAL MODEL TO SUPPORT HEALTHCARE SIMULATION, RESEARCH AND TRAINING"

Virtual

August 2023

University of Washington Simulation Symposium

"ARTIFICIAL INTELLIGENCE AND COMPUTER MODELS FOR SIMULATION"

Seattle, Wa

September, 2023

UW Medicine, Department of Anesthesia Grand Rounds

"DISCUSSION OF ADVANCED OPEN-SOURCE RESEARCH TOOLS TO CONSTRUCT AND EVALUATE THE ADVANCED JOINT AIRWAY MANAGEMENT SYSTEM (AJAMS)"

Seattle, Wa

November 2023

UW BioEngineering 500

"ORGAN SYSTEM MODELING"

Seattle, Wa

May 2024

University of Florida, Systems Medicine Department

"WHOLE BODY PHYSIOLOGY MODEL TO SIMULATE RESPIRATORY DEPRESSION OF FENTANYL AND ASSOCIATED NALOXONE REVERSAL - A MODEL INFORMED REPEAT DOSING ANALYSIS"

Seattle, Wa

June 2024

Pacific Northwest National Laboratory

"BIOGEARS: WHOLE-BODY PHYSIOLOGICAL MODEL TO SUPPORT HEALTHCARE SIMULATION, RESEARCH AND TRAINING"

Seattle, Wa

July 2024

Other Employment

Applied Research Associates, Inc.

Raleigh, NC

BIOMEDICAL MODELING GROUP LEADER (SENIOR ENGINEER, DISTINGUISHED MEMBER OF THE TECHNICAL STAFF)

February 2016 - October 2021

- Lead a multidisciplinary team across 4 different projects
- In charge of agile development processes, product roadmap, delivery scheduling, and direct communication with government customers
- Led and won multiple research and development funds through Defense Health Agency and Army Research Lab grants
- Principal investigator of the BioGears, BurnCare training application, and the traumatic brain injury angiogenesis projects
- Organized teaming across three research hospitals and multiple small businesses
- Communicate research progress through multiple conferences and peer reviewed publications, including the BioGears 2020 conference
- Oversaw implementation of all models associated with BioGears releases 7.0-7.5